Introduction to Frequency Selective Filtering

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Key Concepts

1) Filters are used to separate based on properties of the input.

2) Frequency selective filters separate signals based on the frequencies of the input.

    a) Lowpass filters pass low frequencies and attenuate high frequencies.

    b) Highpass filters pass high frequencies and attenuate low frequencies.

c) Bandpass filters pass mid frequencies and attenuate both high and low frequencies.

d) Bandstop and notch filters pass low and high frequencies and attenuate mid frequencies.

3) Practical filters have some variation in gain in the passband, do not perfectly attenuate in the stop band, and require a range of frequencies to transition between pass and stop bands.

4) A tolerance diagram is used to specify the requirements on a filter magnitude response.

5) Filter phase response is usually chosen in one of three categories:

    a) Linear phase filters only introduce a time delay and do not cause any distortion.

    b) Don't care about the filter response, so there may be phase distortion.

    c) Zero phase, which implies a non causal filter and can only be applied if the complete signal is available.

6) The sampling rate and conversion from continuous- to discrete-time frequency is required if discrete time filter specifications are stated using continuous-time frequency.

Video Screenshots

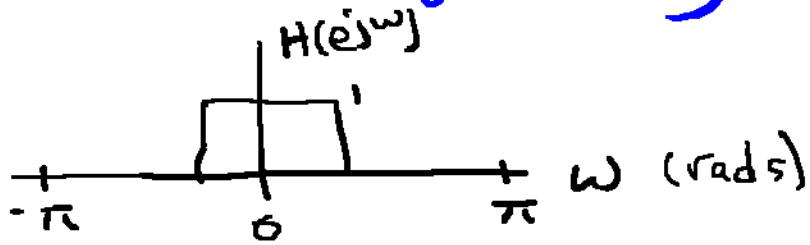
0:04

Filter: to separate

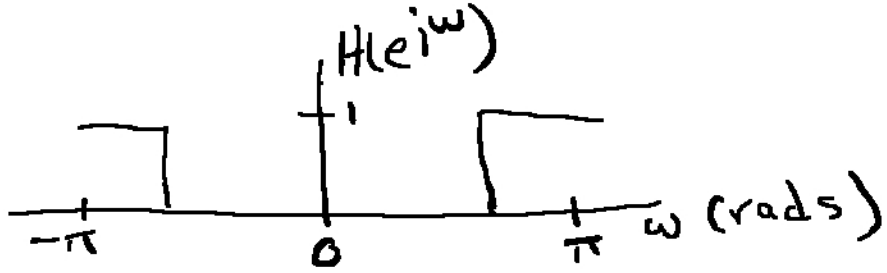
Frequency selective filtering - separate based on frequency



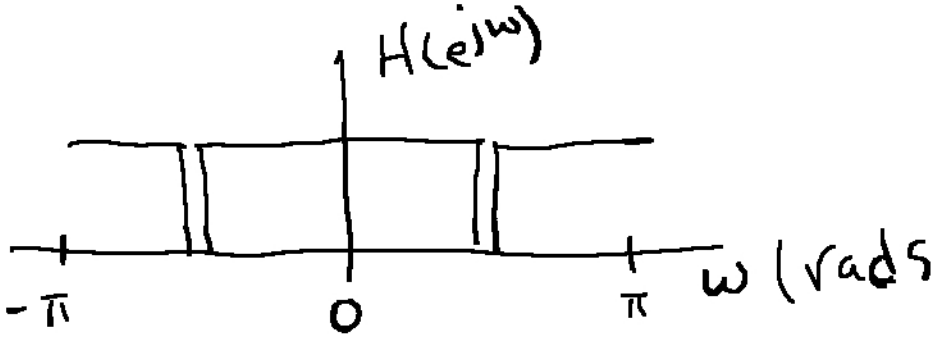
Low pass



Highpass

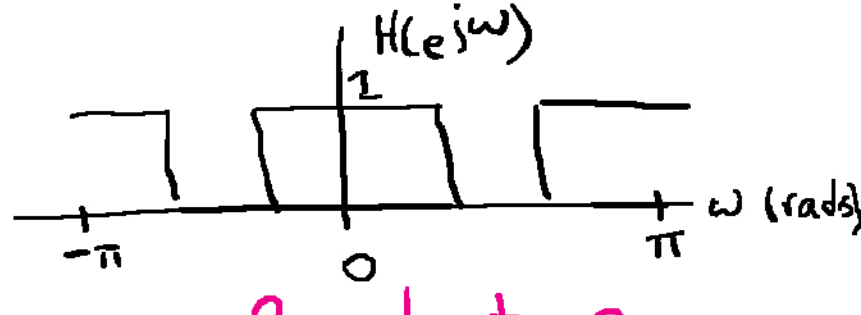


Notch



Bandpass





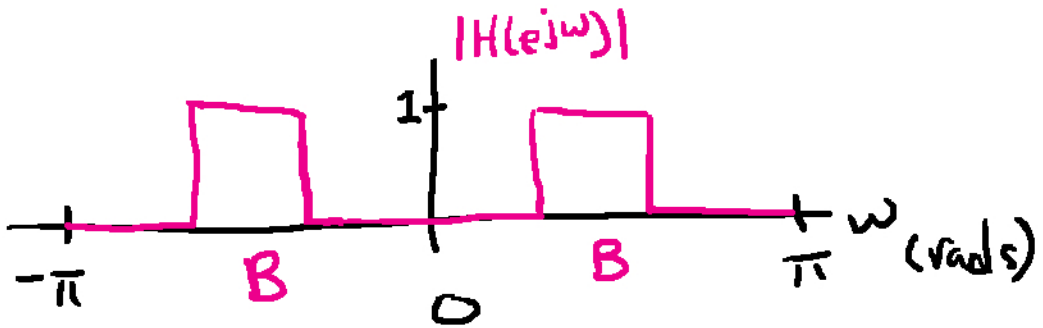
Bandstop

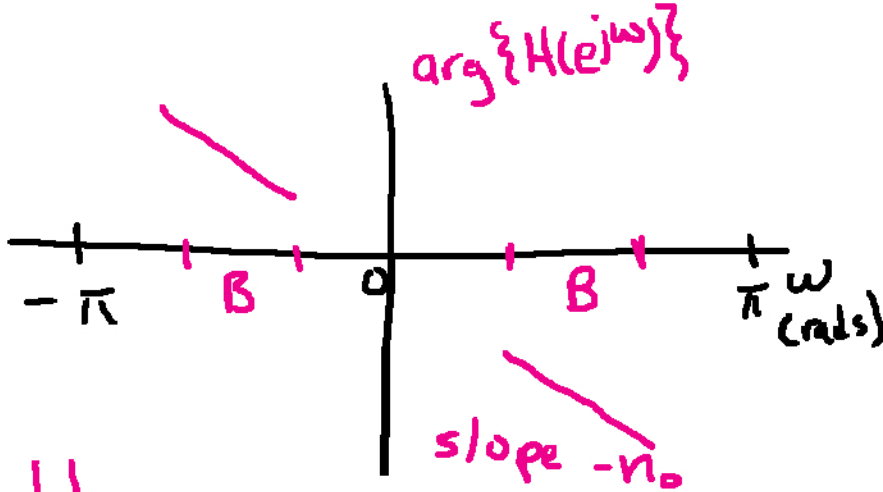
Ideal Filters

$x[n] \in B$ $\quad x[n] \rightarrow \boxed{H} \rightarrow x[n-n_0]$

$y[n] \notin B$ $\quad y[n] \rightarrow \boxed{H} \rightarrow 0$

$x[n-n_0] \overset{DTFT}{\longleftrightarrow} e^{-j\omega n_0} X(e^{j\omega})$



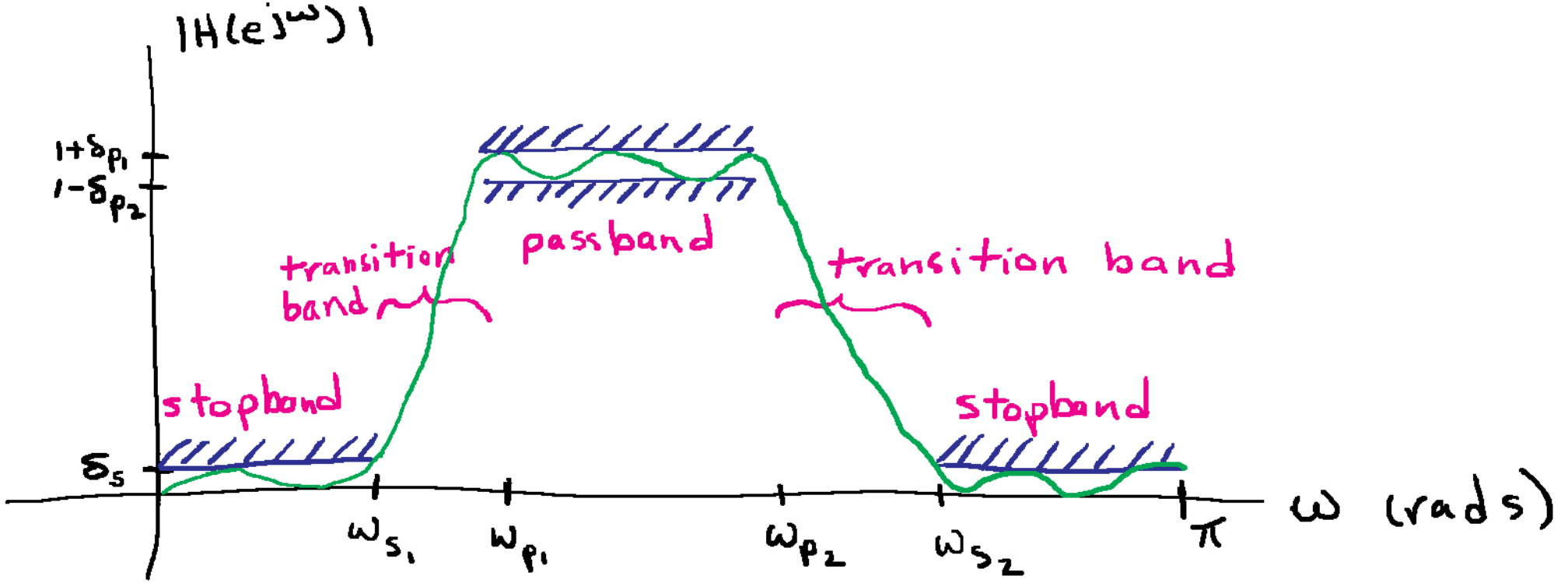


Ideal filters cannot be implemented!

Practical filters: variable gain in passband
nonzero gain in stopband
transition band
nonlinear phase

Filter gain specifications: tolerance diagram

3:39



$|H(e^{j\omega})| < \delta_s$, $|\omega| < \omega_{s_1}$, $\omega_{s_2} < |\omega| < \pi$ stop

$1-\delta_{p_2} < |H(e^{j\omega})| < 1+\delta_{p_1}$, $\omega_{p_1} < |\omega| < \omega_{p_2}$ pass

"don't care", $\omega_{s_1} < |\omega| < \omega_{p_1}$, $\omega_{p_2} < |\omega| < \omega_{s_2}$

Filter phase specifications:

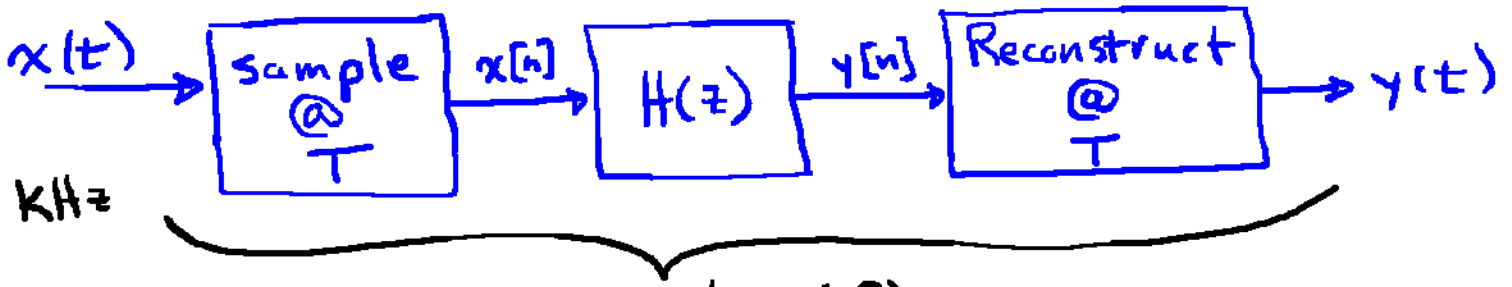
1) Linear – distortionless

2) Don't care

3) Zero-phase – noncausal filtering of stored signals

Example:

7:09



$T = 10^{-4}\,s \Rightarrow f_s = 10\,kHz$

$H_{eff}(f)$

$0.99 < |H_{eff}(f)| < 1$, $f < 2\,kHz$

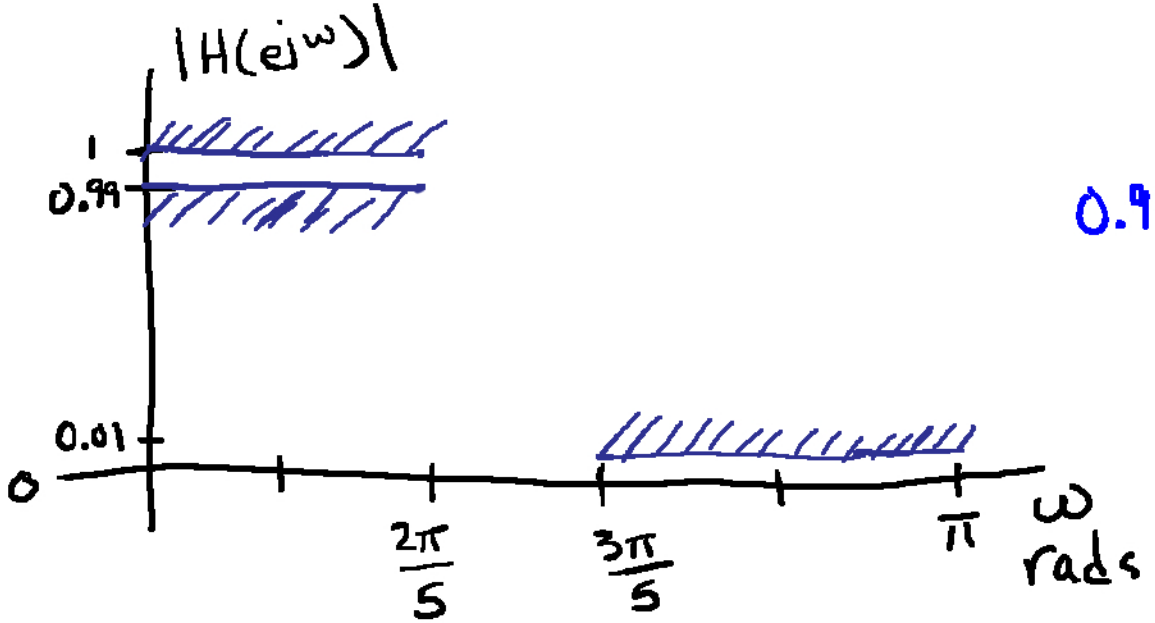
$|H_{eff}(f)| < 0.01$, $f > 3\,kHz$

Find specs on $H(e^{j\omega})$

$\omega = 2\pi f T$

$f = 2kHz \Rightarrow \omega = \frac{2\pi}{5}$

$f = 3kHz \Rightarrow \omega = \frac{3\pi}{5}$



$0.99 < |H(e^{j\omega})| < 1$, $|\omega| < \frac{2\pi}{5}$

$|H(e^{j\omega})| < 0.01$, $|\omega| > \frac{3\pi}{5}$

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